3.10 Use binary numbers to do the following:

3.10.1 Add

3.10.2 Subtract

3.10.3 Multiply

Addition

Binary addition works much like decimal addition. We add the digits in each column, starting from the right and moving left. However, because binary uses only 0 and 1, we must carry whenever the sum in a column is 2 or greater.

Rules for Binary Addition

Addition	Result	Explanation
0 + 0	0	Write 0
0 + 1	1	Write 1
1 + 0	1	Write 1
1 + 1	10	Write 0, carry 1
1 + 1 + 1	11	Write 1, carry 1

Remember: $10_2 = 2_{10}$. When adding 1+1 in binary, the answer is 10_2, not 2.

Note

We can check our answers by converting the binary numbers to decimal.

Examples

Add $1010_2 + 0101_2$

Add $1011_2 + 0110_2$

Add $101_2 + 110_2 + 011_2$

Add $1010_2 + 0011_2$

Add $1110_2 + 0111_2$

Add 101101_2 and 111011_2. Check your answer by converting all three binary numbers to decimal.

A computer adds two unsigned 8-bit binary numbers: 11111111_2 + 00000001_2.

What is the result? How many bits are needed to represent the answer?

Subtraction

Binary subtraction works much like decimal subtraction. We subtract the digits in each column, starting from the right and moving left. Since binary is a base 2 number system, it uses only two digits: 0 and 1. When the top digit is smaller than the bottom digit, we must borrow from the column to the left.

Rules for Binary Subtraction

Subtraction	Result	Explanation
0 – 0	0	Write 0
1 – 0	1	Write 1
1 – 1	0	Write 0
0 – 1	1	Borrow 1 from the next column to the left

Note

When we borrow 1 from the next column, it is worth 10_2 (or 2 in decimal) in the current column.

Therefore, $10_2 - 1_2 = 1_2$.

Examples

Subtract $1010_2 - 0010_2$.

Subtract $1101_2 - 0110_2$.

Borrowing Across Zeros

Sometimes the digit immediately to the left is 0, so we cannot borrow directly from that column. In this situation, we must move farther to the left until we find a 1. We then borrow from that 1 and work back through the zeros.

- Find the nearest 1 to the left.
- Change that 1 to 0.
- Each zero passed through becomes 1 after passing a borrow to the next column.
- The column where borrowing was needed receives 10_2.

Subtract $10000_2 - 00001_2$.

- Start in the rightmost column.
    - We need to subtract 0 - 1, so we must borrow.
- Look at the column immediately to the left.
    - It contains 0, so we cannot borrow directly from it.
    - Continue moving left until we reach the 1 in the leftmost column.
- Borrow from the leftmost 1.
    - The leftmost 1 becomes 0. The borrowed value passes through each zero until it reaches the rightmost column.
    - The three intermediate zeros each become 1 after passing along the borrow.
    - The rightmost column now contains 10_2.
- Complete the subtraction.
    - Rightmost column: $10_2 – 1 = 1$
    - Next three columns: $1 – 0 = 1$
    - Leftmost column: $0 – 0 = 0$

Therefore, $10000_2 - 00001_2 = 01111_2$

Subtract 1110_2 - 0101_2.

Subtract 11010_2 - 01011_2.

Subtract 100000_2 - 001101_2.

Multiplication

Binary multiplication works much like decimal multiplication. We multiply each digit of the second number by the first number, starting from the right.

Since binary uses only the digits 0 and 1, multiplication is relatively straightforward.

Rules for Binary Multiplication

Multiplication	Result
0 × 0	0
0 × 1	0
1 × 0	0
1 × 1	1

Note

- Multiplying a binary number by 0 gives 0.
- Multiplying a binary number by 1 gives the original number.
- When multiplying by a digit farther to the left, shift the partial product one additional position to the left.
- Add the partial products using the rules of binary addition.

Examples

Multiply 1011_2 x 1_2.

Multiply 101_2 x 11_2.

Multiply 1101_2 x 101_2.

Multiply 1101_2 x 1011_2. Check your answer by converting the numbers to decimal.

Explain why multiplying a binary number by 10_2 is equivalent to shifting all its digits one position to the left.